

Personalized Emotion Analysis via Prompting Large Language Models with Writer Information

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Abstract

Personalized emotion analysis aims to account for individual differences in emotional expression by adapting predictions to each writer. Existing personalization methods typically require dedicated feature extraction modules or modifications to model architectures, increasing model complexity. In this paper, we propose a prompt-based personalization framework for emotion analysis that incorporates writer information into the input prompt of a large language model (LLM), without requiring fine-tuning or architectural changes. Specifically, we investigate two complementary types of writer information: personality traits representing stable writer characteristics and post history representing writer-specific patterns of emotional expression. Experimental results on the WRIME corpus in Japanese show that incorporating temporally close post history consistently improves emotion intensity estimation across multiple LLMs. Personality information alone provides limited improvements, but yields additional gains for several models when combined with post history. These findings suggest that prompt-based personalization is a practical alternative to architecture-level personalization for LLM-based emotion analysis.

1 Introduction

Emotion analysis of user-generated text remains a challenging task because emotional expressions are inherently subjective and vary across individuals (Kajiware et al., 2021; Milkowski et al., 2021). The same expression may convey different emotional intensities depending on the writer’s personality, writing style, or habitual patterns of emotional expression (Suzuki et al., 2022b; Lyu et al., 2023). As large language models (LLMs) are increasingly used to analyze user-generated content, modeling such user-specific variation has become an important challenge. Consequently, personalized emotion analysis, which adapts predictions to individual

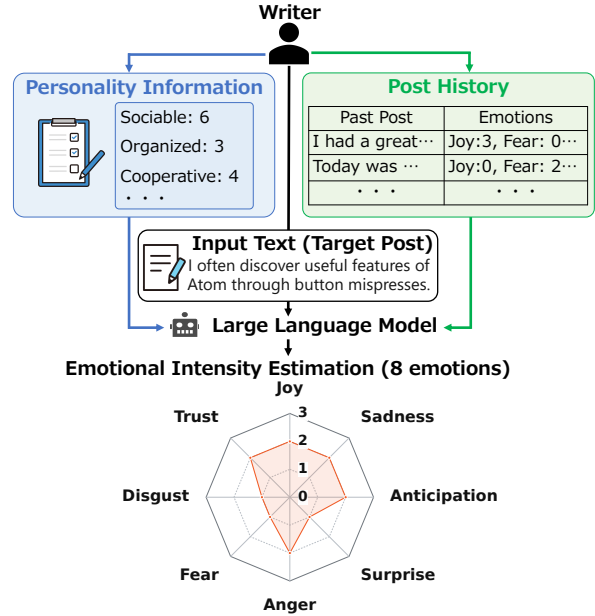


Figure 1: Overview of the proposed method.

writers, has emerged as a promising direction for improving LLM-based emotion analysis.

Previous studies have incorporated writer-specific information, such as personality traits or representations derived from historical posts, into neural emotion analysis models (Suzuki et al., 2022b; Lyu et al., 2023). While these approaches have demonstrated the usefulness of writer information, they typically rely on dedicated user encoders, feature extraction modules, or attention mechanisms beyond the underlying language model (Chen et al., 2016; Wu et al., 2018; Milkowski et al., 2021; Zhang et al., 2024). Such model-specific designs increase architectural complexity and reduce portability across different LLMs.

Recent advances in LLM prompting provide a simple alternative. Instead of modifying model architectures, writer-specific information can be incorporated directly into prompts as natural language context. This naturally raises the following question: Can prompting alone effectively personalize

LLMs for emotion analysis on user-generated text?

In this paper, as shown in Figure 1, we propose a prompt-based personalization framework that injects writer information into LLM prompts without architectural modifications or fine-tuning. We consider two complementary sources of writer information: personality, which represents relatively stable user characteristics, and post history, which captures user-specific patterns of emotional expression and their temporal context. Because both are provided entirely through prompts, the proposed approach can be readily applied across LLMs without introducing dedicated personalization components.

We evaluate the proposed method on WRIME (Kajiware et al., 2021; Suzuki et al., 2022a), a benchmark corpus of Japanese social media posts annotated with writers’ subjective emotional intensities. Experimental results demonstrate that temporally close post history consistently improves emotion estimation across multiple LLMs, while personality information provides complementary gains for several models. These findings suggest that prompting with writer information is an effective, lightweight approach to LLM personalization for user-generated text.

Our contributions are summarized as follows.

- We propose a prompt-based personalization framework that incorporates writer information into LLMs for emotion analysis without requiring model modification or fine-tuning.
- We systematically investigate two complementary forms of writer information—personality and post history—and analyze their respective roles in personalized emotion analysis.
- We demonstrate across multiple LLMs that temporally close post history consistently improves emotion estimation, while personality information provides complementary benefits when combined with post history.

2 Related Work

Early studies on personalized emotion analysis focused on movie and product reviews and proposed methods that combine textual feature representations with representations of writers and products (Tang et al., 2015; Chen et al., 2016; Wu et al., 2018). These studies personalize sentiment polarity classification by learning embedding representations for individual writers and products and integrating them with textual representations through concatenation or attention mechanisms.

Personalized emotion analysis using pretrained language models has also been investigated. Milkowski et al. (2021) combined Personal Emotional Bias (PEB), which represents individual biases in emotion annotations, with textual representations obtained from BERT to perform emotion estimation while accounting for individual differences. In addition, Lyu et al. (2023) proposed a method that constructs user and product representations from past reviews associated with the same user or product and models their relationships using an attention mechanism. By explicitly using past reviews, their method improved sentiment polarity classification performance across multiple pretrained language models.

For personalized emotion analysis using writers’ personality information, Suzuki et al. (2022b) proposed an emotional intensity estimation method that integrates personality assessment results based on the Big Five (Goldberg, 1992) with textual representations obtained from BERT, using the WRIME corpus of Japanese SNS posts. Their experimental results demonstrated that incorporating personality information improves the performance of estimating writers’ subjective emotional intensities.

Recently, methods have also been proposed to efficiently incorporate personalized information into pretrained language models. Zhang et al. (2024) proposed E2LN, which injects user and product information into Layer Normalization and enables both the modeling of long review texts and the incorporation of personalized information through parameter-efficient fine-tuning. Although this method reduces the number of trainable parameters compared with conventional personalization approaches, it still requires introducing a dedicated mechanism for processing personalized information and training the model.

These existing studies require dedicated mechanisms for encoding writer-specific information and integrating it with textual representations, as well as fine-tuning on emotion analysis data. In contrast, we propose a personalization method that directly incorporates personality information and post history into the LLM prompt, eliminating the need for specialized feature extraction mechanisms, modifications to the model architecture, or fine-tuning.

3 Methods

We perform personalized emotion analysis by incorporating writer information into the prompt of

an LLM. Specifically, we consider two complementary forms of writer information: personality, representing stable characteristics of the writer, and post history, capturing writer-specific patterns of emotional expression. The input therefore consists of the target post together with either personality information or post history, or both.

3.1 Personality-based Personalization

The writer’s personality is expected to influence not only how emotions are experienced but also how they are expressed in text. Therefore, we augment the base prompt with personality information corresponding to the writer of the target post and instruct the LLM to estimate emotional intensity while taking the writer’s personality into account.

In this study, we use the Big Five personality model (Goldberg, 1992) to represent personality information. The Big Five model characterizes an individual’s personality along five traits: extraversion, agreeableness, conscientiousness, neuroticism, and openness. This personality information is obtained from a 60-item personality assessment based on the Big Five model (Saito et al., 2001) and is provided for each writer in WRIME. Each item consists of a personality-descriptive word, such as “talkative” or “anxious,” and writers rate how well each word describes them on a seven-point scale. Continuous scores indicating the degree to which each of the five traits applies to the writer are then calculated from these responses.

To investigate the effect of the level of detail in personality information, we use the following two input formats.

- **Personality60:** The response values for the 60 personality assessment items are provided in a format such as “talkative: 6, anxious: 2.” This format provides the LLM with detailed personality information before it is aggregated into the five traits.
- **Big Five:** The five traits—extraversion, agreeableness, conscientiousness, neuroticism, and openness—and their corresponding scores are provided as input. This format presents the writer’s personality to the LLM as a summary along five major dimensions.

3.2 Post History-based Personalization

The relationship between specific expressions and emotional intensities is expected to vary across writers. Therefore, we include multiple posts written

by the same writer and their emotion labels in the prompt as few-shot in-context learning examples, enabling the LLM to capture writer-specific patterns of emotional expression.

We use the following two post-history selection methods.

- **Random:** Ten posts are randomly selected from those written by the same writer.
- **Recent:** Among the posts written by the same writer before the target post, the ten temporally closest posts are selected.

Recent uses only posts available at prediction time, reflecting the writer’s recent emotions and topics. This prevents future information leakage and reflects practical prediction scenarios.

3.3 Combining Personality and Post History

Personality information represents stable characteristics of the writer, whereas post history reflects specific patterns of emotional expression. We therefore incorporate both types of information into the prompt and use them in a complementary manner.

We evaluate **Personality60 + Recent**, which combines the 60-item personality information with post history, and **Big Five + Recent**, which combines the five Big Five traits with post history.

4 Experiments

4.1 Settings

Dataset We evaluate our method using WRIME¹ (Kajiwara et al., 2021; Suzuki et al., 2022a), a Japanese emotion analysis corpus. WRIME contains Japanese SNS posts annotated with writers’ subjective emotions and readers’ objective emotions. We focus on the writers’ subjective emotions and estimate the intensities of Plutchik’s eight basic emotions (Plutchik, 1980): joy, sadness, anticipation, surprise, anger, fear, disgust, and trust. Emotion intensity is represented on a four-point scale, {0, 1, 2, 3}, corresponding to no, weak, medium, and strong, respectively.

For the data split, 7,000 posts from WRIME are used to construct writer-specific information, including post history, and to train the personality feature extractor used in the comparison method (Suzuki et al., 2022b). The remaining posts are divided into 24,000 training instances, 2,000 validation instances, and 2,000 test instances. The

¹<https://github.com/ids-cv/wrime>

Split	Writers	Posts
Feature extraction	60	7,000
Training	40	24,000
Validation	10	2,000
Test	10	2,000

Table 1: Data splits of the WRIME corpus.

training, validation, and test sets contain posts from 40, 10, and 10 writers, respectively, with no overlap in writers across the splits. Table 1 presents the dataset statistics.

Models As a baseline, we use BERT² (Devlin et al., 2019), which was pretrained on Japanese Wikipedia. For the evaluated LLMs, we use LLM-jp³ (LLM-jp, 2024), which was pretrained on Japanese and English data as well as source code, and Swallow⁴⁵ (Fujii et al., 2024; Ma et al., 2025), which was obtained by continual pretraining of an English LLM on Japanese data. We also use Qwen⁶ (Qwen Team, 2026) and Gemma⁷ (Gemma Team, 2026), multilingual models pretrained on text in multiple languages, primarily English.

Comparison Methods We use Japanese BERT as the baseline. For comparison, we employ a basic model that estimates emotional intensity solely from the input post and the personalized method (Personality) proposed by Suzuki et al. (2022b).

In the baseline implementation, the final-layer representation corresponding to the initial special token [CLS] is used as the feature vector for the post, and a fully connected layer predicts one of four emotional intensity labels. We construct an independent classifier for each of the eight basic emotions and fine-tune the pretrained model on the WRIME training data. We set the learning rate to 5.0×10^{-6} , the dropout rate to 0.1, and the batch size to 32, and use AdamW (Loshchilov and Hutter, 2019) for optimization. We also apply early stopping and terminate training when performance

on the validation data does not improve for three consecutive epochs.

In the personalized comparison method (Personality), the responses to the 60-item personality assessment based on the Big Five personality model are transformed into a personality representation. The resulting personality representation is integrated with the textual representation obtained from BERT to estimate emotional intensity while taking the writer’s personality into account.

In the experiments using LLMs, we compare the following settings.

- **0-shot:** Only the instruction and the target post are provided, without writer-specific information or examples with gold labels.
- **10-shot:** Ten posts and their gold emotion labels, randomly selected from the training data without distinguishing writers, are provided as few-shot in-context learning examples.
- **Personality60:** The writer’s responses to the 60-item personality assessment are added to the prompt. Because no examples with gold labels are included, this setting corresponds to a 0-shot setting.
- **Big Five:** The five traits—extraversion, agreeableness, conscientiousness, neuroticism, and openness—and their corresponding scores are added to the prompt. This is also a 0-shot setting because it does not include examples with gold labels.
- **Random:** Ten posts are randomly selected from those written by the same writer as the target post, and the post texts and their gold emotion labels are provided as few-shot in-context learning examples.
- **Recent:** Among the posts written by the same writer before the target post, the ten posts whose timestamps are closest to that of the target post are provided as few-shot in-context learning examples.
- **Personality60 + Recent:** The 60-item personality information and the post history selected by Recent are provided simultaneously.
- **Big Five + Recent:** The five Big Five traits and the post history selected by Recent are provided simultaneously.

²<https://huggingface.co/tohoku-nlp/bert-base-japanese>

³<https://huggingface.co/llm-jp/llm-jp-4-8b-thinking>

⁴<https://huggingface.co/tokyotech-llm/Qwen3-Swallow-8B-SFT-v0.2>

⁵<https://huggingface.co/tokyotech-llm/Qwen3-Swallow-32B-SFT-v0.2>

⁶<https://huggingface.co/Qwen/Qwen3.6-27B>

⁷<https://huggingface.co/google/gemma-4-31B-it>

Model	Method	Joy	Sadness	Anticipation	Surprise	Anger	Fear	Disgust	Trust	Overall
BERT	Fine-tuning	0.524	0.417	0.298	0.286	0.399	0.282	0.412	0.143	0.410
	+ Personality	0.518	0.438	0.315	0.326	0.365	0.294	0.424	0.225	0.424
llm-jp-8B	0-shot	0.564	0.499	0.362	0.339	0.403	0.429	0.354	0.209	0.456
	+ Big Five	0.546	0.507	0.381	0.340	0.357	0.351	0.379	0.214	0.448
	+ Personality60	0.481	0.423	0.341	0.295	0.266	0.265	0.327	0.181	0.376
Swallow-8B	0-shot	0.565	0.453	0.355	0.298	0.350	0.353	0.201	0.221	0.431
	+ Big Five	0.553	0.464	0.423	0.318	0.304	0.368	0.309	0.199	0.435
	+ Personality60	0.521	0.438	0.390	0.306	0.317	0.343	0.271	0.164	0.416
Swallow-32B	0-shot	0.599	0.507	0.385	0.328	0.405	0.382	0.343	0.165	0.464
	+ Big Five	0.569	0.495	0.426	0.323	0.331	0.394	0.375	0.162	0.454
	+ Personality60	0.540	0.493	0.393	0.356	0.354	0.347	0.348	0.210	0.448
Gemma-31B	0-shot	0.676	0.536	0.498	0.494	0.548	0.417	0.477	0.295	0.543
	+ Big Five	0.671	0.540	0.496	0.484	0.525	0.458	0.440	0.269	0.531
	+ Personality60	0.656	0.564	0.470	0.462	0.510	0.458	0.433	0.223	0.513
Qwen-27B	0-shot	0.631	0.468	0.417	0.441	0.519	0.398	0.384	0.180	0.505
	+ Big Five	0.631	0.514	0.427	0.435	0.530	0.418	0.446	0.238	0.518
	+ Personality60	0.611	0.512	0.441	0.459	0.415	0.339	0.411	0.208	0.483

Table 2: Results of writer emotion intensity estimation with personality information. Personality denotes an existing method that incorporates personality information (Suzuki et al., 2022b). Overall denotes the QWK computed over all eight emotions jointly. Bold values indicate the best performance for each model.

The 0-shot and 10-shot settings serve as LLM baselines that do not use writer-specific information. By comparing 10-shot with Random, we examine the effect of selecting examples written by the same writer. By comparing Random with Recent, we further examine the effect of selecting post history while considering temporal proximity.

Metric We use Quadratic Weighted Kappa (QWK) (Cohen, 1968), a standard evaluation metric for ordinal classification tasks.

4.2 Results

4.2.1 Personality-based Personalization

Table 2 presents the evaluation results for personalized methods using personality information. We compare the existing method based on Japanese BERT with three settings for each LLM: the 0-shot setting, which uses only the input post; Big Five, which provides the five Big Five traits; and Personality60, which provides the responses to the 60-item personality assessment.

For BERT, incorporating personality information improved the overall QWK across the eight emotions, confirming the effectiveness of personality information in conventional training-based methods. For the LLMs, however, the effect of personality information varied across models. Big Five improved the overall QWK across the eight emotions for Swallow-8B and Qwen-27B, whereas the

0-shot setting achieved the best performance for the other models. These results suggest that incorporating personality information through prompting is effective for some LLMs, but its effectiveness depends on the characteristics of the model.

A comparison between Big Five and Personality60 shows that Big Five achieved a higher overall QWK across the eight emotions for every LLM. Moreover, Personality60 performed worse than the 0-shot setting for all LLMs. These results suggest that presenting personality information summarized into five major traits may allow LLMs to utilize information relevant to emotion estimation more effectively than directly providing all 60 detailed items. The performance degradation observed with Personality60 may also be attributable to redundant input information and the increased prompt length.

At the emotion level, incorporating the Big Five traits outperformed the 0-shot setting for many models in sadness, anticipation, fear, and disgust. Relatively large improvements were observed for disgust with Swallow-8B and for sadness, disgust, and trust with Qwen-27B. In contrast, no model showed an improvement for joy, and only a limited number of models improved for anger. These findings indicate that the effectiveness of personality information varies across emotions: it is relatively effective for estimating sadness, anticipation, fear, and disgust, but is not necessarily effective for joy or anger.

Model	Method	Joy	Sadness	Anticipation	Surprise	Anger	Fear	Disgust	Trust	Overall
BERT	Fine-tuning + Personality	0.524 0.518	0.417 0.438	0.298 0.315	0.286 0.326	0.399 0.365	0.282 0.294	0.412 0.424	0.143 0.225	0.410 0.424
llm-jp-8B	0-shot	0.564	0.499	0.362	0.339	0.403	0.429	0.354	0.209	0.456
	10-shot	0.587	0.530	0.359	0.347	0.467	0.419	0.352	0.199	0.479
	Random	0.601	0.544	0.369	0.399	0.450	0.473	0.362	0.178	0.489
	Recent	0.618	0.532	0.374	0.412	0.463	0.462	0.380	0.167	0.496
	+ Personality60	0.594	0.540	0.374	0.393	0.447	0.449	0.391	0.176	0.485
	+ Big Five	0.605	0.539	0.391	0.392	0.462	0.469	0.386	0.201	0.496
Swallow-8B	0-shot	0.565	0.453	0.355	0.298	0.350	0.353	0.201	0.221	0.431
	10-shot	0.532	0.495	0.291	0.269	0.479	0.365	0.200	0.083	0.413
	Random	0.530	0.523	0.334	0.274	0.433	0.427	0.282	0.078	0.422
	Recent	0.598	0.513	0.328	0.363	0.506	0.477	0.405	0.101	0.471
	+ Personality60	0.588	0.511	0.321	0.329	0.462	0.477	0.379	0.116	0.459
	+ Big Five	0.594	0.504	0.307	0.351	0.498	0.500	0.393	0.094	0.465
Swallow-32B	0-shot	0.599	0.507	0.385	0.328	0.405	0.382	0.343	0.165	0.464
	10-shot	0.603	0.535	0.381	0.340	0.428	0.430	0.289	0.164	0.476
	Random	0.619	0.537	0.408	0.366	0.429	0.441	0.209	0.143	0.478
	Recent	0.635	0.519	0.350	0.409	0.467	0.456	0.359	0.132	0.490
	+ Personality60	0.586	0.544	0.333	0.392	0.448	0.417	0.309	0.118	0.465
	+ Big Five	0.614	0.537	0.340	0.419	0.438	0.443	0.314	0.138	0.479
Gemma-31B	0-shot	0.676	0.536	0.498	0.494	0.548	0.417	0.477	0.295	0.543
	10-shot	0.667	0.529	0.431	0.497	0.603	0.410	0.494	0.247	0.530
	Random	0.671	0.534	0.461	0.489	0.572	0.446	0.525	0.206	0.535
	Recent	0.695	0.540	0.462	0.523	0.584	0.443	0.559	0.304	0.564
	+ Personality60	0.696	0.548	0.491	0.517	0.586	0.460	0.560	0.300	0.570
	+ Big Five	0.697	0.540	0.497	0.510	0.599	0.471	0.569	0.311	0.572
Qwen-27B	0-shot	0.631	0.468	0.417	0.441	0.519	0.398	0.384	0.180	0.505
	10-shot	0.639	0.493	0.399	0.440	0.512	0.403	0.373	0.115	0.498
	Random	0.647	0.480	0.410	0.415	0.511	0.430	0.373	0.122	0.496
	Recent	0.672	0.495	0.414	0.452	0.574	0.423	0.411	0.095	0.513
	+ Personality60	0.665	0.536	0.441	0.473	0.537	0.460	0.448	0.159	0.526
	+ Big Five	0.667	0.520	0.419	0.455	0.558	0.452	0.432	0.151	0.521
Annotator	A	0.490	0.314	0.377	0.339	0.255	0.354	0.305	0.135	0.367
	B	0.590	0.436	0.438	0.400	0.285	0.363	0.448	0.118	0.449
	C	0.553	0.485	0.426	0.347	0.302	0.365	0.275	0.131	0.426

Table 3: Results of writer emotion intensity estimation using writer information in terms of QWK. Personality denotes an existing method that incorporates personality information (Suzuki et al., 2022b). Overall denotes the QWK computed over all eight emotions jointly. Bold values indicate the best performance for each model.

4.2.2 Post History-based Personalization

Table 3 presents the results of methods using post history, methods combining post history with personality information, and human evaluation. The annotator scores represent the agreement between each reader’s objective emotion annotations and the writer’s subjective emotion labels in terms of QWK, without access to the writer’s personality information or post history. For each LLM, we compare 0-shot, 10-shot with examples randomly selected from the training data, Random with examples randomly selected from posts written by the same writer, and Recent with temporally close posts written before the input post. We further combine Recent with either Personality60 or Big Five to investigate the complementary use of post history and personality information.

Random outperformed the standard 10-shot set-

ting for many models. This result indicates that using posts written by the same writer as few-shot examples enables LLMs to capture writer-specific emotional expressions and writing styles. Furthermore, Recent outperformed 0-shot for all LLMs and achieved the highest overall QWK across the eight emotions for several models. This suggests that, even among examples written by the same writer, selecting temporally close posts written before the input post effectively reflects the writer’s recent emotional expressions and context. The comparison with the standard 10-shot setting also confirms that, in addition to the number of examples, the correspondence between the examples and the writer as well as their temporal proximity are important in few-shot learning.

At the emotion level, Recent improved performance for a wide range of emotions, including joy,

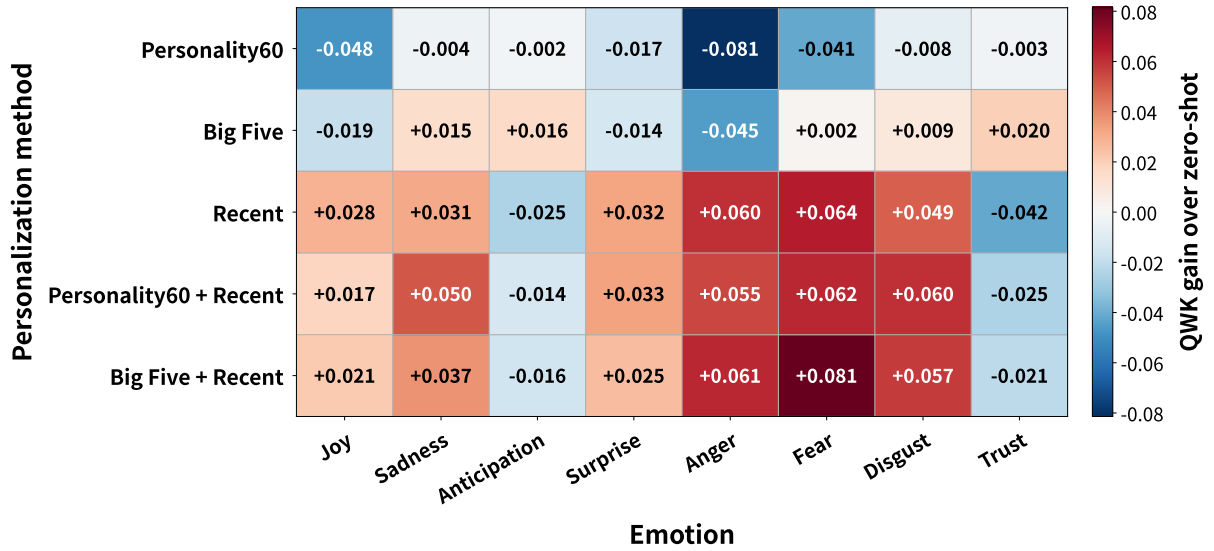


Figure 2: Emotion-wise QWK improvements achieved by each personalization method, averaged across the evaluated LLMs. Improvements are calculated relative to the 0-shot setting for each model.

sadness, surprise, anger, fear, and disgust. In particular, compared with Random, Recent achieved higher performance for all models on joy, surprise, and disgust. These results suggest that temporally close post history is useful not only for capturing writer-specific emotional expressions but also for distinguishing emotions relevant to the target post.

When post history was combined with personality information, the overall QWK across the eight emotions exceeded that of Recent alone for Gemma-31B and Qwen-27B. In particular, the combination with Big Five performed well for Gemma-31B, whereas the combination with Personality60 performed well for Qwen-27B. This may be because post history captures the writer’s recent emotional expressions and specific expressive tendencies, while personality information complements them by representing the writer’s relatively stable characteristics. Thus, combining post history with personality information enables the model to capture writer-specific characteristics from different perspectives and yields additional improvements that cannot be obtained from post history alone.

Compared with human evaluation, the best overall QWK across the eight emotions achieved by each LLM exceeded the highest score among the human annotators. This result demonstrates that the proposed method can achieve high estimation performance for writers’ subjective emotional intensities, which are difficult for readers to infer, by leveraging post history and personality information as additional cues. Overall, incorporating tempo-

rally close post history from the same writer into the prompt consistently improves performance across multiple LLMs. Moreover, combining post history with personality information enables the model to capture the writer’s stable characteristics and temporally varying patterns of emotional expression in a complementary manner.

5 Analysis

5.1 Effectiveness across Emotions

In this section, we analyze which emotions benefit from personality information and post history using emotion-wise QWK. Figure 2 shows the QWK improvement for each emotion, averaged across the evaluated LLMs. The improvement is calculated by subtracting the 0-shot QWK from that of each personalization method. A positive value indicates improvement over the 0-shot setting, whereas a negative value indicates performance degradation. As shown in Figure 2, personality information and post history exhibited different patterns in the emotions they improved.

When only Personality60 was provided, performance decreased relative to the 0-shot setting for many emotions. In particular, the QWK for anger decreased by approximately 0.08 on average. In contrast, Big Five yielded slight improvements for sadness, anticipation, and trust, although the overall gains were small.

Recent, which uses post history, produced substantial improvements for anger, fear, and disgust. In particular, the QWK increased by approximately

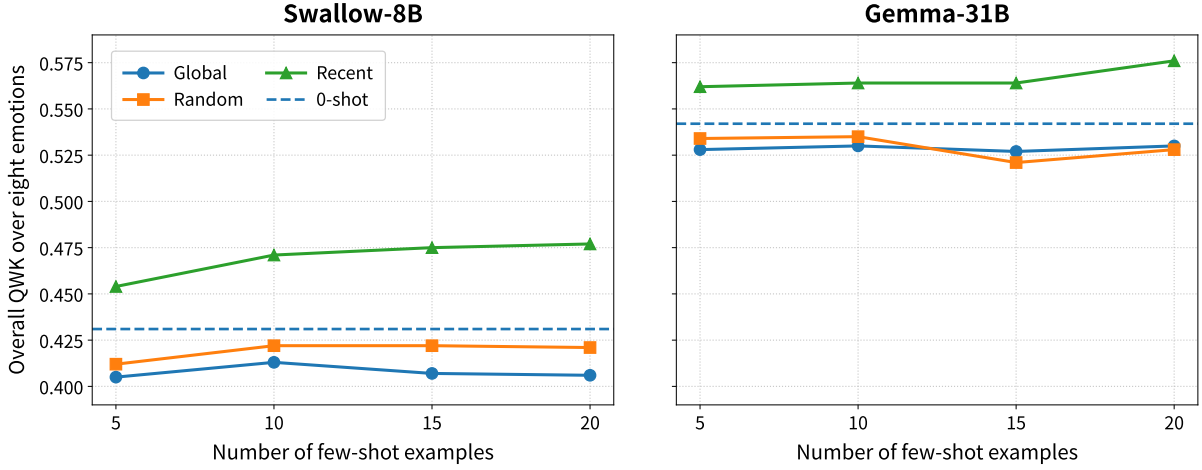


Figure 3: Changes in the Overall QWK across eight emotions with different numbers and selection strategies of few-shot examples. Global denotes training examples selected without considering the writer, Random denotes posts randomly selected from the same writer, and Recent denotes temporally close posts from the same writer that precede the input post. Dashed lines indicate the performance of the 0-shot setting for each model.

0.06 for fear, 0.06 for anger, and 0.05 for disgust. In contrast, no improvement was observed for anticipation or trust, indicating that the effectiveness of post history varies across emotions.

When personality information was combined with post history, the methods maintained large gains for anger, fear, and disgust, as with Recent alone, while further improving sadness. In particular, Big Five + Recent achieved the largest gain of approximately 0.08 for fear and approximately 0.06 for both anger and disgust. In contrast, the gain for trust was small, and some settings underperformed the 0-shot setting.

Overall, these results indicate that post history is particularly effective for estimating anger, fear, and disgust, for which writer-specific patterns of emotional expression are likely to be reflected. Although personality information alone provides only limited improvements, combining it with post history yields additional gains for some emotions, particularly sadness.

5.2 Effect of Few-shot Example Size

To analyze the effects of the number of examples and the example selection method on performance, we conducted experiments using Swallow-8B and Gemma-31B while varying the number of few-shot examples among 5, 10, 15, and 20. We considered three example selection methods: Global, which selects examples from the training data without distinguishing writers; Random, which randomly selects posts written by the same writer; and Recent, which selects temporally close posts written before

the target post. The results are shown in Figure 3.

For Swallow-8B, Global and Random underperformed 0-shot at all example sizes, whereas Recent consistently improved performance from 0.454 with 5 examples to 0.477 with 20. This indicates that temporal relevance is more important than simply increasing the number of few-shot examples.

For Gemma-31B, Global and Random also underperformed 0-shot, while Recent consistently surpassed it. Its performance remained stable from 5 to 15 examples and increased to 0.576 with 20, suggesting model-dependent gains from additional history.

Overall, these results show that few-shot performance depends more on example relevance than quantity. Temporally close posts from the same writer outperformed 0-shot even with few examples and provided further gains as the number increased.

6 Conclusion

We addressed writers’ subjective emotion intensity estimation and proposed a personalized method that incorporates personality information and post history into the LLM prompt. Experiments on WRIME showed that temporally close post history consistently improved performance, whereas personality information alone had limited effects. Combining both yielded further gains, demonstrating the effectiveness of prompt-based personalization for LLM emotion analysis.

Acknowledgments

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Limitations

Our experiments are limited to the WRIME corpus, which contains Japanese SNS posts, and the generalizability of the findings to other languages and domains remains to be investigated. In addition, the available post history is limited, and the effectiveness of the proposed method may depend on the number and temporal distribution of posts available for each writer. Furthermore, our method assumes that emotion labels for past posts are available at prediction time. In practical applications, these labels may need to be collected through self-reports or estimated automatically.

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